

Output for the 3D Distance, 3D Velocity, and Mass of the 100th disk particle of the Milky Way at SnapNumber 0

Screenshot:

```
Distance_Mag = np.around((np.sqrt((x**2 + y**2 + z**2))*u.kpc),3)

Velocity_Mag = np.around((np.sqrt((vx**2 + vy**2 + vz**2))*u.km/u.s),3)

Mass = (mass*10**10)*u.M_sun

return Distance_Mag, Velocity_Mag, Mass

ParticleInfo('MW_000-Copy1.txt',3,15)
```

```
[4]: (<Quantity 4.282 kpc>, <Quantity 507.466 km / s>, <Quantity 1000500. solMass>)
```

So

3D Distance = 4.282 kpc

3D Velocity = 505.466 km/s

Mass = 1000500 Solar Mass

Conversion of the 3D Distance of the particle from kpc to lightyears

Screenshot:

```
return Distance_Mag, Velocity_Mag, Mass

#ParticleInfo('MW_000-Copy1.txt',1,1)
Distance_Mag, Velocity_Mag, Mass = ParticleInfo('MW_000-Copy1.txt',2,99)
print(np.around(Distance_Mag.to(u.lyr),3))

13845.338 lyr
```

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So the 3D Distance is converted to 13845.338 lightyears